Rasheed Abdul Haq Kozhiparamban

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Professional Summary

- Senior Machine Learning Engineer with a **Ph.D. in Artificial Intelligence** and **8+ years across AI research and applied ML**, specializing in industrial computer vision, time-series forecasting, virtual sensing, and enterprise generative AI.
- Experience across the ML lifecycle: data quality, model development, validation, deployment, **MLOps**, and human review. Hands-on with Python, PyTorch, TensorFlow, OpenCV, Azure ML, and AWS Bedrock.
- Co-inventor on **three granted Indian patents** and co-author of **five journal papers**, including IEEE and Springer publications.

Education

- **Ph.D.** in Artificial Intelligence (Deep Learning, Predictive Modeling & Time-Series Forecasting), National Institute of Technology (NIT), Puducherry, India, 2018 – 2024
- **M.Tech** in Electronics Design Technology, National Institute of Electronics & Information Technology (NIELIT), Calicut, India, 2012 – 2014
- **B.Tech** in Electronics and Communication Engineering, School of Engineering, CUSAT, Cochin, India, 2005 – 2009

Experience

Chevron Engine, Bengaluru, India: Senior Machine Learning Engineer
July 2025 – Present

- Led the AI/ML handover of a **production drone-inspection platform** to an in-house team, defining data-readiness and knowledge-transfer criteria and supporting end-to-end **user acceptance testing (UAT)** through sign-off.
- Designed a two-stage component-segmentation and corrosion-detection pipeline to suppress background false positives; aligned independently versioned **Azure ML** endpoints, model registration, and inference orchestration. Supported production monitoring and serverless architecture simplification.
- Built high-resolution inspection workflows with overlapping tiles, prediction stitching, configurable class thresholds, and structured outputs; merged detections into continuous annotations for inspector review.
- Improved utility-asset segmentation through dataset audits, class-imbalance analysis, label refinement, and preprocessing reviews; established annotation standards and subject matter expert (SME) validation gates.
- Delivered worker detection and contributed to safety video analytics covering activity recognition, personal protective equipment (PPE), camera placement, and occlusion; packaged a **YOLO11 fire-and-smoke model** for edge testing.
- Built a physics-based **thermal tank-level estimation prototype** using radiometric calibration and temperature-gradient analysis; delivered a web demo with weak-signal handling. Ground-truth validation remains pending.
- Contributed sensor-data integration and annotated inspection outputs to **digital twin** workflows for condition monitoring and planning; advised on change-detection methods, user-triggered inference, and production cost and licensing considerations.
- Evaluated and shortlisted computer vision vendors for environmental, health, and safety (EHS), assessing accuracy, latency, hazardous-area certifications (ATEX/IECEx), edge readiness, privacy, and workflow integration.
- Authored an annotation handbook and architecture/training references; co-authored CV deployment guidelines on risk-based metrics, verification, and lifecycle governance. Led planning, knowledge transfer, stakeholder demos, SME reviews, and structured Senior ML Engineer interviews.

SONY, Bengaluru, India: Lead AI/ML Engineer
April 2025 – June 2025 (3-month contract engagement)

- Built secure **RAG pipelines** using AWS Bedrock and Claude with enterprise data from **SharePoint and Confluence** via MCP, with source citations and retrieval restricted to enterprise knowledge sources.
- Integrated and tuned **Jira/Confluence MCP clients**, using AWS Secrets Manager for token-based access and validating source scope parameters.
- Developed **LLM tools for task analysis** and automated risk scanners, streamlining deployments across Windows and AWS EC2 environments.

DIMAAG-AI, Bengaluru, Karnataka, India: Senior Software Engineer AI/ML
February 2023 – February 2025 (2 years 1 month)

- Developed production computer vision pipelines for industrial and agricultural applications using **YOLOv8**, segmentation models (U-Net, DeepLabV3, PSPNet), and vision-language models.
- Implemented real-time action recognition with **SlowFast**; validated camera field of view using feature matching, optical flow, and statistical comparisons.
- Processed **3D point clouds** for segmentation and feature extraction; applied Random Forest, Gradient Boosting, and MLP classifiers to structured prediction tasks.
- Improved dataset quality through image clustering, augmentation, statistical analysis, geometric transformations, and hyperparameter tuning.
- Built PDF retrieval and **RAG** workflows with FAISS, Pinecone, and LangChain, including structured extraction, SOP summarization, multi-step LLM workflows, and role-based access.
- Used **Azure ML** and Azure AI services for training and deployment, with **MLflow** for experiment tracking and model versioning; led experiments and mentored junior engineers.

National Institute of Technology, Puducherry, India: Senior Research Fellow
July 2018 – January 2023 (4 years 7 months)

- Developed **virtual sensing** and multivariate time-series forecasting for aquaculture water quality using LSTM, GRU, bidirectional, hybrid CNN-LSTM/CNN-GRU, and attention-based models.
- Evaluated forecasting accuracy and computational efficiency through hyperparameter optimization and experiments on aquaculture datasets.
- Built an **IoT-based low-cost monitoring system** and dissolved-oxygen estimation models using sensor data to reduce reliance on expensive physical sensors.
- Co-authored peer-reviewed publications and co-invented patented virtual-sensing and hybrid prediction methods; resulting publications and grants are listed below.

Patents:

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF", Indian Patent No.: 397595, Date of Grant: May 25, 2022.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING", Indian Patent No.: 552671, Date of Grant: October 21, 2024.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION", Indian Patent No.: 579506, Date of Grant: January 30, 2026.

Publications:

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. "Deep Learning Enhanced Virtual Sensing: A Cost-Effective Approach for Water Quality Monitoring in Aquaculture" *IEEE Transactions on AgriFood Electronics*, June 2025.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models", *IEEE Access*, vol. 10, pp. 60078-60098, June 2022.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction System based on Adam Optimised LSTM Neural Network for Aquaculture: A Case Study in Kerala, India", *J. Inst. Eng. India Ser. B*, *Springer*, 103, 2177–2188, October 2022.
4. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Accurate Dissolved Oxygen Prediction for Aquaculture Using Stacked Ensemble Machine Learning Model", *Natl. Acad. Sci. Lett.*, *Springer*, 46, 203–207, January 2023.
5. Rahul Gandh D, Harigovindan V P, Rasheed Abdul Haq K P, and Amrtha Bhide, "Attention-driven LSTM and GRU deep learning techniques for precise water quality prediction in smart aquaculture", *Aquaculture International*, vol. 32, pp. 8455–8478, 2024. doi:10.1007/s10499-024-01574-5.

MEA Engineering College, Perinthalmanna, Kerala, India: Assistant Professor, Department of ECE
June 2015 – July 2018 (3 years 1 month)

- Taught undergraduate and postgraduate electronics courses, including Digital Signal Processing, Signals & Systems, Digital System Design, and Communication Systems; developed course and laboratory materials for the KTU curriculum transition.
- Mentored students, guided embedded/wireless projects, coordinated seminars and project assessments, and served as an external laboratory examiner. Built spreadsheet-based attendance and academic-performance tracking tools.

Way in Electronic Solutions, Calicut, India: Founder & Senior Embedded Software Engineer
August 2014 – June 2017 (2 years 11 months; part-time alongside teaching from June 2015)

- Co-founded an embedded systems startup and led requirements, architecture, firmware, hardware integration, testing, deployment, and customer support for industrial automation and monitoring products.
- Developed parking guidance, ultrasound probe testing, security, digital signage, and animal-health monitoring solutions using **Embedded C**, 8051, Raspberry Pi, Linux, RS-485/UART, and sensor interfaces.
- Built web monitoring interfaces with PHP, MySQL, JavaScript, and HTML/CSS; authored design documents, coding standards, technical proposals, and customer presentations.

M. N. Dastur & Co. (P) Ltd., Bengaluru, India: DSP Intern (Embedded Image Processing)
September 2013 – July 2014 (11 months)

- Developed an **Automatic Number Plate Recognition (ANPR)** pipeline for Indian vehicles using MATLAB, OpenCV, and C++, covering plate localization, character segmentation, OCR, and SVM classification.
- Implemented image preprocessing and geometric correction, then ported and evaluated the pipeline on Ubuntu Linux and an ARM-based **BeagleBoard-xM**; documented prototype results and deployment challenges.

Bharti Airtel Ltd., Kerala, India: Network Surveillance Engineer
January 2010 – February 2012 (2 years 2 months)

- Monitored telecom network health, service availability, alarms, and KPIs through OMC-R tools; investigated faults and coordinated root-cause analysis with field engineers and vendors.
- Managed alarm escalations, planned outages, preventive maintenance, and remote troubleshooting; analyzed trends and produced operational reliability reports.

Personal AI Projects

Arogyamitra — AI medical triage assistant for Telegram (Kerala, India): **2025**
LLM-driven chatbot that collects demographics and symptoms, triages risk, and recommends specialists.

- Delivered production-ready Telegram bot with structured multi-field demographic and symptom collection.
- Implemented multi-provider LLM fallback and persistent rate-limit tracking for robust uptime.
- Built precise doctor-matching with exclude-keyword logic to prevent misclassification.

Stack: Python, python-telegram-bot, Groq (LLM), OpenRouter/Gemini, HuggingFace, CSV (data), Railway (deployment)
Links: GitHub — Demo

Skills

- **Programming Languages:** Python, C++, MATLAB
- **AI/ML Expertise:** Predictive Modeling, Time-Series Forecasting, Statistical Learning, Machine Learning, Deep Learning, Computer Vision, Thermal Image Analysis, Instance Segmentation, Change Detection, Vision-Language Models, Digital Twins, NLP, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Agentic AI
- **Libraries/Frameworks:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, YOLOv8/YOLO11, PyTorchVideo, Hugging Face Transformers, LangChain, NumPy, Pandas, Matplotlib, NLTK, spaCy
- **MLOps & Cloud:** Azure ML, Managed Endpoints, Model Registry, Multi-model Orchestration, Azure Functions, Azure AI Services, AWS Bedrock, MLflow, Git
- **Vector Databases & Retrieval:** FAISS, Pinecone
- **Operating Systems:** Linux, Windows
- **Professional Skills:** Technical Leadership, Cross-functional Collaboration, Technical Mentoring, Stakeholder Communication, Annotation Standards, Model Verification, UAT, Technical Documentation